

بیوشیمی

1- All the following enzymes are important in gluconeogenesis EXCEPT:

PEPCK

Fructose 1,6 biphosphatase

Glucose 6 phosphatase

✓ Phosphorylase

2- TCA cycle enzymes are located in..... ?

Cytosol

✓ Mitochondria

Nucleus

Cell membrane

3- Which of the following enzymes is important in glycolysis?

✓ Enolase

Pyruvate carboxylase

Malate dehydrogenase

Glycerokinase

4- How many ATP molecules are synthesized after complete oxidation of glucose?

26-28

✓ 30-32

34-36

38-40

5- Which of the following is synthesized in HMP shunt?

NAD

NADH

FADH₂

✓ NADPH

6- Some patients with G6PD deficiency develop hemolytic anemia if they?

- ✓ Produce low amounts of NADPH
- Are treated with oxidant drug
- Ingest corn
- Contract a severe infection

7- Which of the following enzymes is important in glycogenolysis?

- Phosphatase
- Phosphokinase
- ✓ Phosphorylase
- Phosphodiesterase

8- All the Following are anaerobic fermentation EXCEPT:

- Conversion of pyruvate to lactate
- Conversion of pyruvate to ethanol
- ✓ Conversion of pyruvate to acetyl CoA
- Synthesis of acetaldehyde

9- Ammonia produced in muscle tissue is transported to liver mainly in the form of?

- ✓ Alanine
- Glutamate
- Ammonia
- Pyruvate

10- Which glucose transporter is active in muscle tissues?

- GLUT-1
- GLUT-2
- GLUT-3
- ✓ GLUT-4

11- Which of the following IS NOT a the Kreb's cycle intermediate?

Succinate



Glucoronate

Oxaloacetate

Cirtrate

12- The end product of glycolysis is -----?



Pyruvic Acid

Glucose

Citric acid

Glycogen

13- Which of the following IS NOT an NPN compound in the blood?



Glutamine

Uric acid

Urea

Creatinine

14- The first nucleotide formed during pyrimidine de novo synthetic pathway is -----?

PRPP

Uric acid



OMP

IMP

15- The final product of purine catabolism in birds and reptiles is -----?

Bilirubin

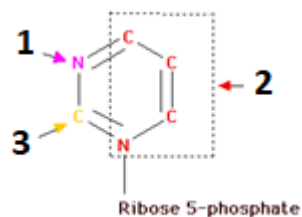
Uric acid



Allantoin

creatinine

16- In the structure below, the numbered groups arise from1.....,2....., and3.....?



- Glutamine, Aspartate, CO₂
- CO₂, Aspartate, Glycine
- Aspartate, CO₂, Glutamine
- Glutamine, Glycine, Aspartate

17- Inosine is made from the addition of ribose to which of the following nitrogenous bases?

- Adenine
- Uracil
- Cytosine
- Hypoxanthine



18- All of the following may lead to hyperuricemia (GOUT), EXCEPT?

- Cancer chemotherapy
- ADA deficiency
- HG-PRT deficiency
- Lesch-Nyhan syndrome



19- What types of molecules serve as the starting materials for synthesis of URIC ACID?

- Amino acids
- Carbohydrates
- Purine bases
- Complex lipids such as lecithin and cholesterol



20- Which enzyme catalyzes the final stage of purine base catabolism?

Guanine deaminase



Xanthine oxidase

Nucleotidase

Adenosine deaminase

21- A child has been diagnosed with severe combined immune deficiency. Which enzyme is defective is?

Purine nucleoside phosphorylase



Adenosine deaminase

Guanase

Xanthine oxidase

22- Tophus is a clinical sign of which condition?

Hypouricemia

Hyperuremia



Hyperuricemia

Dehydration

23- β -Aminoisobutyric is the end product of degradation of which of the following:

Cytosine

Uracil

Guanine



Thymine

24- In severe combined immune deficiency (SCID) which of the following increases and inhibits the synthesis of deoxyribonucleotides by ribonucleotide reductase, causing problems with the immune system (death of lymphocytes, immunodeficiency disease)?

- ✓ dATP
- ATP
- AMP
- ADP

25- In Lesch-Nyhan syndrome

- ✓ There is a defect or lack in the HGPRT enzyme
- There is an excess in the HGPRT enzyme
- HGPRT enzyme is not involved in the disease
- Ammonia into ammonia buffer

26- All the following organs are involved in creatine formation, EXCEPT.....?

- Liver
- Muscle
- ✓ Brain
- kidney

27- A 2-week-old child was brought to the emergency. The parents had realized that the child's diapers would turn black if kept unwashed for a longer time. Which amino acid pathway is implicated in this phenomenon?

- ✓ Tyrosine
- Leucine
- Histidine
- Threonine

28- The best endogenous substance for calculating the estimated GFR is

Ammonia

Urea

Creatine

✓ Creatinine

29- Azotemia is the same as

Hyperuricemia

✓ Uremia

Hyperammonemia

Hyperbilirubinemia

30- Which of the following is a zymogen?

✓ Proelastase

Carboxypeptidase

Chymotrypsin

Trypsin

31- Hartnup disease is due to a problem in the absorption of

Cystine

✓ Phenylalanine

Cysteine

Glycine

32- Which of the following amino acids is only ketogenic (lipogenic)?

✓ Lysine

Arginine

Serine

Tyrosine

33- Lack of arginase in urea cycle gives rise to which of the following conditions?

Citrullinemia

Hyperammonemia

✓ Argininemia

Argininosuccinate aciduria

34- Transamination reactions require the cofactor

Thiamine

Riboflavin

✓ Pyridoxine

Niacine

35- A positive nitrogen balance occurs

✓ During pregnancy

During starvation

During surgery

In kwashiorker

36- During urea synthesis, the 2 nitrogen atoms in urea are contributed by

✓ Ammonia and aspartate

Glutamine and glutamate

Ammonia and glutamate

Ammonia and alanine

37- Hemolytic anemia causes

✓ Prehepatic bilirubinemia

Hepatic bilirubinemia

Posthepatic bilirubinemia

Icterus

38- Which of the following compounds has the most positive effect on insulin secretion?

- ✓ Glucose
- Amino acids
- Lipids
- Epinephrine

39- All of the following reactions occur in the fasting state, EXCEPT?

- Glucagon stimulates gluconeogenesis in the liver
- ✓ Epinephrine stimulates glycogenesis in the liver
- Glucagon stimulates proteolysis in muscles
- Insulin is released in basal amounts from pancreatic cells

40- All the following provide carbon compounds necessary for the synthesis of new glucose in the fasting state, EXCEPT?

- Glycerol
- Amino acids
- Lactate
- ✓ Ketone bodies

41- Which of the following statements about insulin-induced glucose entrance into tissues IS CORRECT?

- Glucose uptake in intestinal epithelial cells is a passive and insulin-dependent transport
- The entry of glucose into skeletal muscle cells is an active and insulin-dependent transport
- The entry of glucose into the liver of diabetics is impaired
- ✓ Glucose uptake from renal tubules is not affected in diabetics

42- Which of the following is correct about LONG-TERM FASTING?

- ✓ Muscles use ketone bodies to produce energy
- The liver uses the ketone bodies of the kidneys to produce energy
- Adipose tissue produces the ketone bodies
- The heart tissue uses the glucose produced by the gluconeogenesis pathway to produce energy

43- All of the following are correct, EXCEPT?

Chylomicrons carry dietary Lipids

VLDL deliver the endogenously derived lipids to the peripheral tissues

Chylomicrons are first released into the lymph and then enter the blood

✓ Carbohydrate maldigestion or malabsorption causes steatorrhea

44- All of the following are correct about TAG oxidation, EXCEPT?

β -oxidation occurs in the mitochondrial matrix

✓ Hormone-sensitive lipase (HSL) is activated by insulin to lyse TAG

Carnitine transports the long-chain acyl group from the cytosol into the mitochondrial matrix

The glycerol released during TAG degradation cannot be metabolized by adipocytes

45- The chemiosmotic hypothesis involves all of the following, EXCEPT?

A mitochondrial membrane is impermeable to protons

Electron transport by the respiratory chain pumps protons out of inner membrane of mitochondria

Proton flow in to the mitochondria depends on the presence of ADP and P_i

✓ Only proton transport is strictly regulated, other positively charged ions can diffuse freely across the mitochondrial membrane

46- A thermogenin would most likely lead to:

Inability to oxidise NADH by the electron transport chain

Inability to produce heat by the operation of the electron transport chain

✓ Operation of the electron transport chain without ATP production

A dependence on ADP for the electron transport chain to operate

47- All of following about production of ketone bodies are correct, EXCEPT?

- ✓ Ketoacidosis is seen in cases of high fat diet feeding
- A fruity odor on the breath, which results from increased production of acetone
- A symptom of uncontrolled type 1 diabetes mellitus (T1D)
- Ketoacidosis may also be seen in cases of excessive ethanol consumption

48- All of the following about TAG digestion in the stomach are correct, EXCEPT?

- Catalyzed by lingual and gastric lipase
- ✓ Hydrolyze FA from TAG molecules long-chain length (>18 carbons)
- Help individuals with pancreatic insufficiency
- Important role in lipid digestion in infants (milk fat is the primary source of calories)

49- The key regulatory enzyme of bile acids synthesis is.....

- HMG Co-A reductase
- HMG Co-A Synthase
- HMG Co-A Lyase
- ✓ 7- α Hydroxylase

50- When excess acetyl-CoA, produced by the liver, cannot be used by the TCA cycle, it is accumulated in the body as....

- Glucose
- Triglycerides
- ✓ Hydroxybutarate
- Acetyl-CoA

51- Which of the following is not non-functional plasma enzyme?

Alkaline phosphatase

Aspartate aminotransferases

✓ Lipoprotein lipase

Creatine kinase

52- Serum acid phosphatase level increases in:

✓ Paget disease

Acute pancreatitis

Hepatobiliary disease

Cirrhosis

53- Which of the following is not a clinical condition associated with lactate dehydrogenase?

Acute myocardial infarction

✓ Cholecystitis

Hepatic disorders

Hemolytic anemia